TPC cosmic-test analysis
User and modification guide

User and developer reference: Python and C++/ROOT

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This manual covers detector mapping, raw inspection, event displays, manual cleaning, and the independent voltage study. Configuration excerpts, command help and example figures are refreshed by the manual builder. All commands below run from the workspace root, unless an absolute path is supplied. The original input ROOT files are read-only.

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1 Workspace and requirements

Location	Purpose
`data/root/`	Input files, named `ntupleNNNN.root`.
`analysis/`	Working Python interface: `tpc.py`, `detector.py`, `voltage.py`.
`analysis_root/`	Native ROOT interface: `TPC.C` and `TPCDetector.h`. No Python is called.
`config/`	Shared detector cuts, independent study cuts, 1D binning and 2D pair definitions.
`results/python/`, `results/root/`	Outputs, grouped by workflow and run.
`docs/`	This LaTeX source, PDF, example figures, generated excerpts and builder.
`bundles/`	Shareable source bundles. Carry the sibling `config/` and `docs/` folders when sharing. Input data are not bundled.

ROOT requires C++17. Python requires PyROOT, NumPy and Matplotlib; interactive Python windows also require a working TkAgg backend. File-only Python exports use `--batch`. ROOT uses `-b` for batch mode; omit `-q` when you want an interactive ROOT session to remain open. PDF rebuilding additionally requires Python and `pdflatex` with the standard packages used in this source.

2 Quick start: inspect one run

The normal inspection opens both groups together: a 1D summary and seven individual 1D canvases, plus a 2D summary and four individual 2D canvases (13 windows for the present branch set). Data are read once per combined invocation. Python uses one shared mapping and selection before entering one window loop.

```
root -l 'analysis_root/TPC.C()'                      # menu
root -l 'analysis_root/TPC.C("raw",1616)'            # 1D + 2D
root -l 'analysis_root/TPC.C("raw --1d",1616)'
root -l 'analysis_root/TPC.C("raw --2d",1616)'
root -l 'analysis_root/TPC.C("raw-clean",1616)'      # both, cleaned

python analysis/tpc.py                               # menu
python analysis/tpc.py raw 1616                      # both
python analysis/tpc.py raw 1616 --1d
python analysis/tpc.py raw 1616 --2d
python analysis/tpc.py raw 1616 --apply-cuts
python analysis/tpc.py raw 1616 --batch               # save, no windows
```

ROOT dimension flags belong *inside the quoted command string.* Python also accepts `--dimensions both|1d|2d` (or `--dim`). Raw inspection bypasses all manual cuts and automatic event vetoes. Cleaned inspection uses the detector database and that run's event-veto ledger. Mapping applies in both. The old `raw2d` / `raw2d-clean` commands remain 2D-only shortcuts.

2.1 Focus and export locations

```
root -l 'analysis_root/TPC.C("raw:Channel",1616)'
root -l 'analysis_root/TPC.C("raw2d:Charge_vs_Pad",1616)'
python analysis/tpc.py raw 1616 --branches Channel
python analysis/tpc.py raw 1616 --plots Charge_vs_Pad
python analysis/tpc.py raw 1616 --1d --branches Time --range 80 100
```

Focused exports use `branches_NAME/` or `plots_NAME/` subfolders so that they preserve full-run outputs. Specify `--dimensions both` in Python when combining focused selections from both groups. Both engines save ordinary outputs in `results/ENGINE/raw/runNNNN/`; cleaned outputs add `cleaned/`, and 2D outputs add `2d/`.

3 Command and option reference

The menus provide the same operations as the command interface. ROOT arguments are `TPC(command,run,event,inputDirectory,outputDirectory,vetoCSV,cutsDB,studyCutsDB)`. Empty strings retain defaults. Dimension flags belong inside the ROOT command string.

ROOT command	Python command	Effect
raw	raw RUN	Uncut 1D and 2D inspection together; saves files and opens windows.
raw --1d / raw --2d	raw RUN --1d / --2d	Select one inspection group; --both explicitly requests both.
raw-clean	raw RUN --apply-cuts	Both groups with detector DB cuts and existing automatic veto IDs applied.
raw-clean --1d / --2d	raw RUN --apply-cuts --1d / --2d	Cleaned inspection of one group.
raw:Channel (or another field)	raw RUN --branches Channel	Focused 1D original-channel diagnostic. raw:Pad uses mapped pads.
raw2d:NAME	raw RUN --plots NAME	Focused 2D pair from the pair CSV.
raw2d / raw2d-clean	raw2d RUN [--apply-cuts]	Backwards-compatible 2D-only shortcuts.
cuts	cuts RUN	List active manual rules only; does not apply them or open plots.
event / event3d	event RUN [--mode 3d]	Browse the present physical event display with detector cleaning and vetoes.
save	event RUN --save	Export an event. ROOT exports both modes; Python exports the selected mode.
examples	examples	Export established run1615/event42535 and run1616/event5584 displays. Other input files are needed if not bundled.
voltage	voltage	Independent voltage comparison; freshly classifies event noise, then applies detector and study cuts.
info	info	Report paths, mapping and configuration.
check	check	Configuration/mapping regression checks; no data-cleaning plots.

Python option	Meaning
--batch	Save inspection outputs without opening windows. In ROOT use -b; use -q only when automatic exit is wanted.
--dimensions both|1d|2d	Select inspection groups; --dim is an alias. --1d, --2d and --both are short forms.
--branches FIELD [FIELD ...]	Focus on named 1D quantities; alone it selects 1D.
--plots NAME [NAME ...]	Focus on configured 2D pairs; alone it selects 2D.
--bins N / --bin-width W	Override 1D bin count or width (mutually exclusive).
--range LOW HIGH	Set 1D plotting edges; does not create a cut.
--log-y / --linear-y	Override the 1D count-axis scale. 2D colour scaling uses LogZ in its CSV.
--binning-file PATH	Alternate base 1D axis CSV, also inherited by 2D axes.
--plots-file PATH	Alternate 2D pair/axis CSV.
--input-dir / --output-dir	Select input directory and output root; see generated help for command-specific layout.
--pad-map PATH	Alternate Python mapping table. ROOT uses its auxiliary table.
--cuts-db PATH	Complete replacement detector DB, including global rules; used only for cleaned data/displays.
--veto-file PATH	Explicit precomputed run/event veto ledger.
--event ID	Event to browse/export; otherwise the first eligible event is used.
--save / --mode 2d|3d	Event export and display mode.
--example RUN:EVENT	Repeat to select Python examples.
--study-cuts-db PATH	Additional independent voltage-study restrictions only.
--voltage-file PATH	Alternate run/voltage assignments.
--voltage-override RUN=VOLTS	Resolve an ambiguous assignment; repeatable.
--exclude-run RUN	Exclude a run from voltage overlays.
--noise-channels / --module-channels / --module-synchronous	Override noise detection thresholds during Python voltage processing; regenerate the ledger afterwards.

Use --help on the Python command for its full syntax. The appendix contains generated help from the current implementation. ROOT path/DB overrides are positional, as shown in the files section.

4 Data and detector mapping

Each input row is a recorded pulse. Multiple rows can belong to the same event or channel. The tree is named `ntuple`. Branches used by the analysis are:

`Event`	Stored event identifier; a pulse count is not an event count.
`Channel`	Original zero-based ROOT electronics channel.
`Hit`	Stored pulse index; the 2D Hit view uses this value, not the total pads in an event.
`Time`	Stored time value in native units.
`Charge`	Stored charge/amplitude value in native units; no energy calibration is assumed.
`Pedestal`	Stored pedestal value.
`Blockhead`	Additional stored field, deliberately omitted from inspection plots.

The established wiring table is `auxiliary/pad_map.csv` in each engine's folder. Printed fADC

labels start at 1; **fADC = ROOT Channel + 1**. Pad numbers start at 0, increase left to right and bottom to top, and continue from medium pads 0–249 to large pads 250–409. Examples: fADC153 = Channel152 = Pad0; fADC16 = Channel15 = Pad24; fADC505 = Channel504 = Pad250.

Mapping is a coordinate conversion, independent of cuts. Unconnected channels have `Pad=-1`. Their hits are retained in flow cells of Pad histograms. Other raw distributions, including the separate Channel diagnostic and time/charge correlations, retain those hits. `pad_map_used.csv` records the wiring used. **Do not add or subtract 1 from a Pad number.**

5 Detector-cleaning database

Edit `config/run_cuts.ini`. It contains only common `[global]` and per-run `[run NNNN]` blocks. Write conditions for values to *reject*:

```
[global]
reject = Time >= 228
reject = Channel in 271,149

[run 1616]
# Illustrations only: enable after inspecting this run.
# reject = Channel == 152
# reject = Event in 1000..1020
# reject = Pad == 42 && Time >= 90 && Time < 95
# reject = Charge > 3500
```

Separate lines are OR: a hit matching any rejection line is removed. Conditions joined by `&&` on one line are AND. Operators are `== != < <= > >=`, `in`, and `not in`. Lists use commas; `in 40..45` includes both endpoints. A histogram interval with edges 90 and 95 is `Time >= 90 && Time < 95`. A leading `#` disables a line and also starts an inline comment. There are no arbitrary functions, parentheses, or `||`; add another rejection line for OR. Invalid syntax and missing branches produce explicit errors.

5.1 Channels, fADC and pads in cuts

`Channel == 152`, `FADC == 153`, and `Pad == 0` are equivalent ways to reject the same connected pad. One is sufficient. Rejections act on hit rows before populating *all* cleaned plots, including mapped Pad plots. A pure Event rule removes every matching event's row; a Channel/Time rule removes only matching pulses. Fields are case-insensitive and also support Hit, Charge, Pedestal and Blockhead. Numeric thresholds may be signed, fractional or scientific notation.

```
python analysis/tpc.py cuts 1616
root -l 'analysis_root/TPC.C("cuts",1616)'
```

These commands **list rules only**; they do not open plots or run an analysis. To apply the listed detector rules and display both groups, run:

```
root -l 'analysis_root/TPC.C("raw-clean",1616)'
python analysis/tpc.py raw 1616 --apply-cuts
```

Keep any custom input/DB path overrides in the plotting command. Event displays apply the detector rules automatically. Uncut `raw` bypasses them. Reopen/re-run the command after DB edits. Uncut inspection intentionally ignores the DB. The current detector database is reproduced in Appendix A.

5.2 Automatic event-noise vetoes: detection and application

An automatic veto is an event-level cut: every pulse with that run/event ID is removed when vetoes are enabled. The original ROOT file is never modified. Detection and application are separate steps.

What is detected. The classifier uses all original pulses and all electronics channels, including unconnected ones, before Time/Channel/Charge cuts or pad mapping. Let D be the number of distinct channels in an event. The current setup has eight electronics modules, with module number $\lfloor \text{Channel}/72 \rfloor$. For each module define C as its number of distinct channels, H as its total pulse rows, and S as the largest number of distinct channels sharing exactly the same stored Time value. Multiple pulses on one channel do not increase D, C or the synchronous-channel count S.

With the present settings, veto the event if *either*:

1. $D \geq 288$ (detector-wide occupancy); or
2. any module has $C \geq 58$ and also either $S \geq 50$ or $H \geq 3C$ (module flooding with exact-time synchrony or repeated pulses).

There is no time-window tolerance in S: equality is on the stored Time value. A module is an electronics group, not a physical pad row. These are heuristic noise criteria; a veto is a classification under these thresholds, not a fitted physics track decision. The settings keys are noise_channels, module_channels, module_synchronous, module_hits_per_channel, channels_per_module and readout_modules. Each engine reads its own settings.csv. Changing one copy does not change the other. The current settings are reproduced in the appendix.

When the veto is applied.

Command/workflow	Automatic event veto behaviour
ROOT raw / raw --1d / raw --2d; Python raw without --apply-cuts; raw2d shortcuts	**OFF**. All raw pulses remain, including flooded events. No manual cuts are applied either.
ROOT raw-clean; Python raw --apply-cuts (1D or 2D)	**ON, using the existing veto ledger**. Entire matching events are removed in addition to detector DB cuts.
Event/event3d, save and examples	**ON, using the existing ledger**, before selecting the first passing pulse per channel for display.
Independent voltage study	**Detects afresh** from each available listed run's full input, applies the resulting vetoes and writes a new ledger. Its separate study hit cuts do not restrict the noise-classifier input.
cuts	Lists manual DB rules only. It neither detects noise nor opens data plots.

Thus, **a raw plot is deliberately not noise-cleaned**. This allows examination of the events that would be removed. To see the effect, compare raw and raw-clean with identical binning. Mapping remains active in both; mapping does not itself remove noisy events.

Where the veto IDs come from. The ledger is keyed by `(run,event)`, never event number alone. Python prefers `results/python/voltage/rejected_events.csv`; ROOT prefers the

equivalent under its configured output root. Otherwise the engine uses its bundled auxiliary ledger (Python also accepts the former legacy layout). An explicit veto override takes precedence. CSV evidence includes distinct-channel occupancy, total pulses, detector-wide flags and per-module flooding evidence.

Cleaned inspection does not rerun the automatic classifier. It reads the IDs already present in this ledger. If a run has no rows in the ledger, no automatic IDs are vetoed for that run; an empty run entry is not evidence that a new dataset was scanned. If the ledger is missing, cleaning raises an error. For new data, regenerate the ledger using the voltage workflow with the relevant run listed, or supply a validated run/event CSV using --veto-file (Python) or vetoCSV (ROOT). After changing noise thresholds, regenerate the ledger: changing settings alone cannot change IDs in an existing file. Explicitly override the ledger when unrelated input files reuse the same run numbers. Ordinary detector DB edits take effect on the next cleaned invocation; they do not regenerate the automatic ledger.

6 One-dimensional histograms and annotations

The 3-by-2 summary shows Event, Pad, Hit, Time, Charge and Pedestal. Each has an individual canvas. Original Channel (0–575) is an additional independent diagnostic, including unconnected inputs. All histograms count **Raw Hits**.

Underflow counts values below the lower edge; **Overflow** counts values at or above the upper edge. Every bin uses ROOT's half-open interval $[\text{low}, \text{high})$, including the last bin. **Hits on unconnected channels** counts selected pulses with no assigned pad. Those hits contribute to Pad underflow but appear at their actual Channel coordinates in the electronics plot. Counts have thousands separators.

Edit `raw_binning.csv`: Branch, BinWidth, Bins, Min, Max, LogY. Empty bounds use the uncut extent. Bins takes precedence over BinWidth. Min/Max are plotting edges, not cuts. Raw and cleaned versions use the same axis extent. Python's `--bins`, `--bin-width`, `--range`, and `--log-y`/`--linear-y` override the 1D view for one call.

7 Two-dimensional histograms

A 2-by-2 summary and one individual canvas per configured pair show correlations. The colour scale counts Raw Hits, with logarithmic colour enabled by LogZ. Empty cells are white. X/Y axes remain linear. All accepted pulses contribute; there is no first-hit-per-channel reduction.

The following table is generated from the *current* pair file, including the configured orientation and overrides. Empty options inherit 1D binning.

Name	X	Y	Overrides
Event_vs_Pad	Pad	Event	YBinWidth=64, LogZ=true
Hit_vs_Pad	Pad	Hit	LogZ=true
Charge_vs_Pad	Pad	Charge	LogZ=true
Charge_vs_Time	Time	Charge	LogZ=true

Edit `raw_2d_plots.csv` to change/add pairs. X/Y define horizontal/vertical fields. XBinWidth, XBins, XMin, XMax and corresponding Y options override each axis. An explicit count takes precedence over a width. The plot Name controls filenames and focused selection and must be unique and filename-safe. Python `--plots-file` chooses an alternate pair file.

For individual-event resolution in a narrower region, set YBinWidth=1, YMin=8000, YMax=9000 on an Event-on-Y row. These edges do not change the DB. A four-million-cell limit prevents accidental enormous dense histograms; narrow the range or increase bin width when needed. The default event grouping in the shipped file was chosen for a manageable overview; the current file always takes precedence.

Outside plotted ranges counts each hit once, including hits outside both axes. ROOT retains all flow cells and corners; per-axis flow totals can overlap. Sparse CSVs contain all nonzero cells, including flows: bin 0 is underflow, N+1 is overflow, and open-end bounds are infinite. A correlation with no Pad axis retains unconnected-channel hits in its ordinary cells if within the plotted limits.

8 Event displays

Development status: the event display is the current, evolving visualization implementation. Its layout and hit-selection policy are subject to further development. It is not a completed spatial track-reconstruction result. Physical event displays place both pad sections according to the established geometry. They apply detector cuts and run-local vetoes, then retain the *first passing pulse per channel* for display. Later good pulses remain eligible if an earlier pulse was rejected. This reduction is exclusive to displays, not raw/voltage histograms.

```
root -l 'analysis_root/TPC.C("event",1616,5584)'
root -l 'analysis_root/TPC.C("event3d",1616,5584)'
root -l -b -q 'analysis_root/TPC.C("save",1616,5584)'
root -l -b -q 'analysis_root/TPC.C("examples")'

python analysis/tpc.py event 1616 --event 5584
python analysis/tpc.py event 1616 --event 5584 --save
python analysis/tpc.py event 1616 --event 5584 --mode 3d --save
python analysis/tpc.py examples
```

ROOT save exports charge/time views in 2D and 3D, exact displayed-hit CSVs and ROOT objects including the `display_hits` tree. Python event save exports PNG/PDF for the chosen mode; examples also export a CSV of exact displayed hits. Outputs live under `results/ENGINE/events/runNNNN/`; comparisons have a separate subfolder. Missing pad values are drawn at zero; that does not establish that a pad is dead. Display charge scales start at zero, so negative values may be visually clipped. Time heights are native Time, not a calibrated drift coordinate.

9 Independent voltage study

This workflow uses detector cleaning from `run_cuts.ini` and then adds `voltage_study_cuts.ini`. Only this study opens that second file. Its channel-range restriction is independent of normal raw/cleaned/event inspection.

```
root -l -b -q 'analysis_root/TPC.C("voltage")'
python analysis/tpc.py voltage
python analysis/tpc.py voltage --study-cuts-db config/voltage_study_cuts.ini
```

The run/voltage table is `data/datafiles-voltage.txt` for Python and the ROOT auxiliary copy for ROOT. Conflicting voltage assignments are excluded until corrected (Python also has `--voltage-override RUN=VOLTS`). Repeated voltages select the eligible file with most distinct

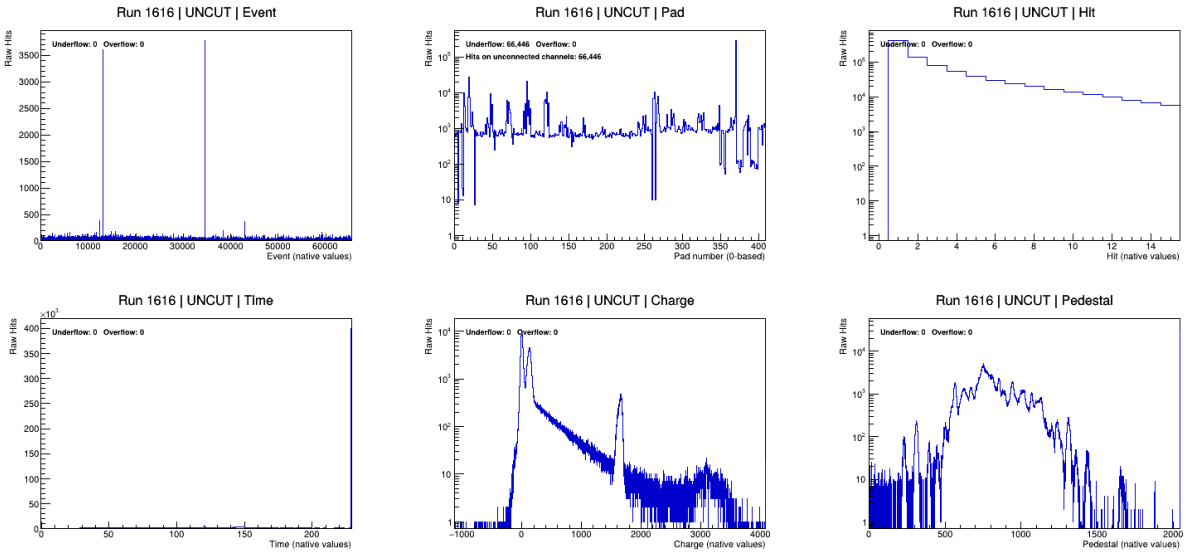


Figure 1: Run 1616 uncut 1D summary; generated using the current ROOT palette and settings.

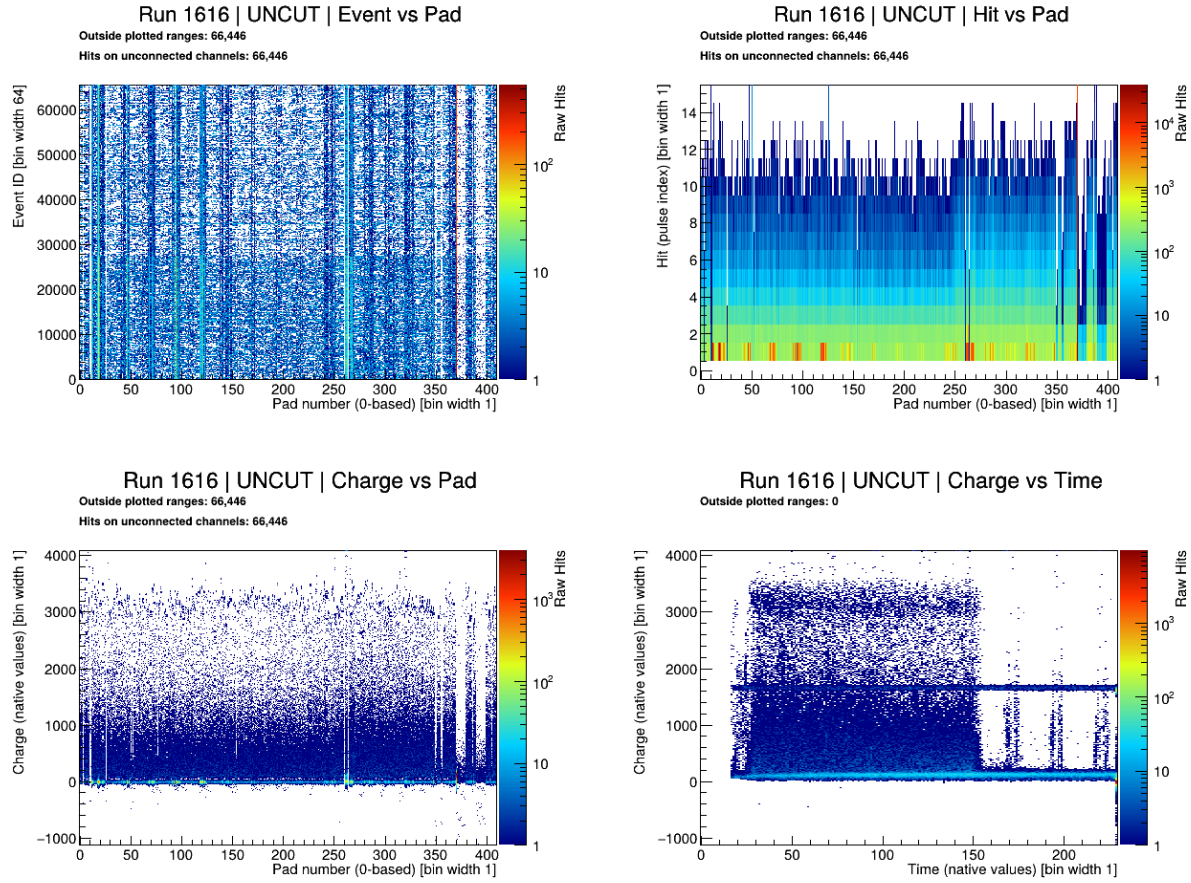


Figure 2: Run 1616 uncut 2D summary. Axis orientation and resolution come from the current pair CSV.

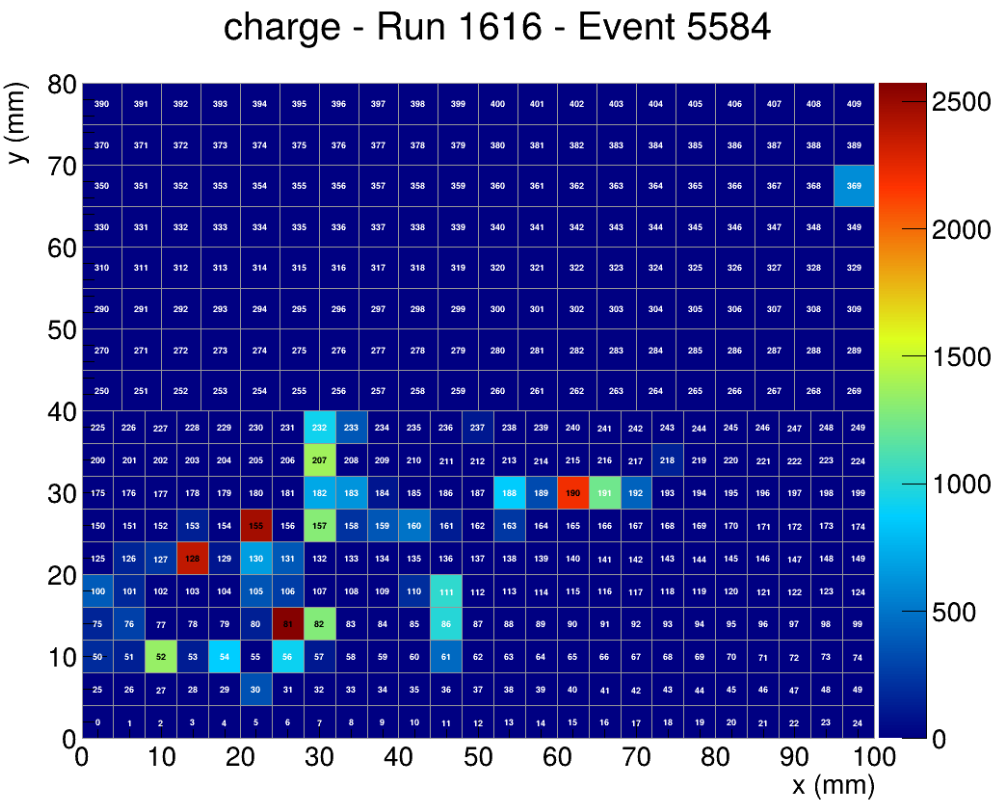


Figure 3: Run 1616, event 5584: numbered physical charge display.

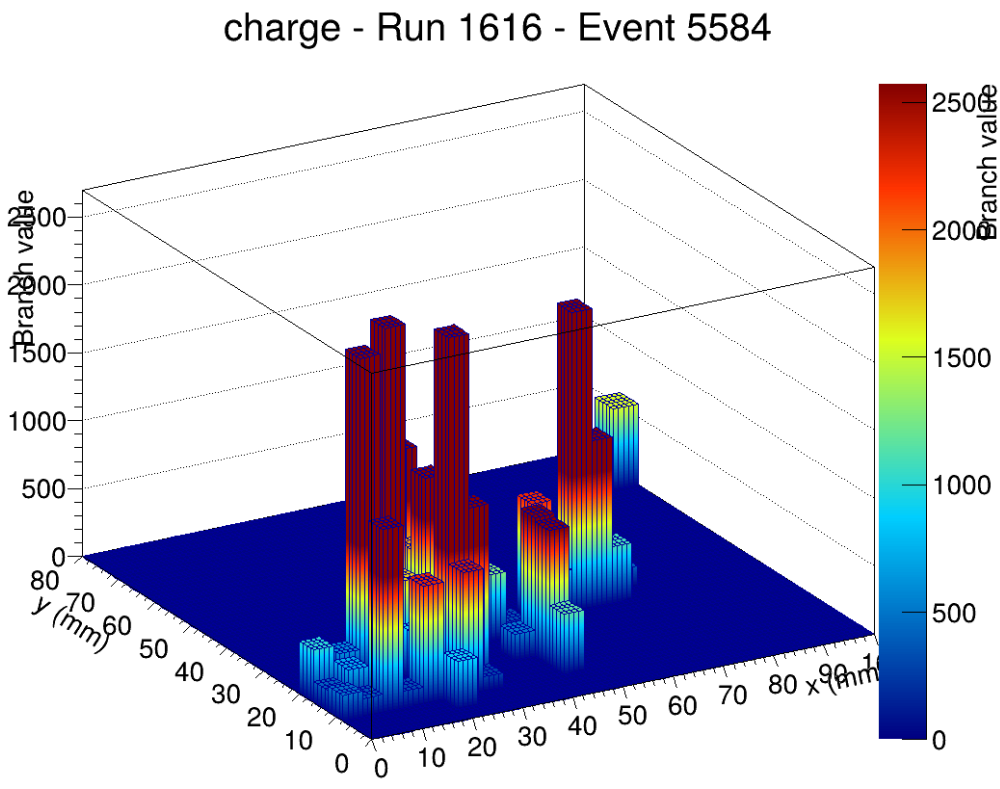


Figure 4: The same event in 3D. Heights are displayed values; no track reconstruction is performed.

events before cuts; ties use clean event count and then the lower run ID. Excluded scan runs are configured in settings.

Count overlays preserve accepted pulse totals. Unit-area overlays compare shapes by dividing by accepted-hit count. Neither is a live-time normalization. Histograms count all accepted pulses. The occupancy/audit plots explain detector-wide and module-flooding vetoes. Output goes to `results/ENGINE/voltage/`, including histograms, run summaries and rejected-event evidence. Both parsed cut files are snapshotted separately. Extend the voltage histogram's `time_max` setting if you expand the accepted Time range.

10 Files, reproducibility and custom locations

Raw 1D files: `raw_histograms.root`, overview/individual PNG/PDFs, per-quantity bin CSVs and summary (Python JSON, ROOT CSV). Raw 2D files: `2d/raw_2d_histograms.root`, overview/individual PNG/PDFs, sparse bin CSVs and summary. Both carry native editable histograms and TCanvas objects. Reopen a ROOT file and use `new TBrowser;` to inspect its keys. Clean exports include `cuts_used.ini`; the voltage study additionally saves `voltage_study_cuts_used.ini`. Snapshots describe the version actually used, so older exports can legitimately carry older configuration text.

```
root -l results/root/raw/run1616/raw_histograms.root
// At the ROOT prompt:
new TBrowser;

python analysis/tpc.py raw 1616 --input-dir /path/to/rootfiles --output-dir /path/to/
    output
python analysis/tpc.py raw 1616 --apply-cuts --cuts-db /path/to/cuts.ini --veto-file /
    path/to/veto.csv
```

An alternate detector DB completely replaces the default, including global cuts. The ROOT interface has the following optional positional parameters; empty strings retain defaults:

```
TPC(command, run, event, inputDirectory, outputDirectory,
    vetoCSV, cutsDB, studyCutsDB);
// Example export with alternate input and detector DB:
TPC("save",1616,5584,"/path/to/data","/path/to/output","","/path/to/cuts.ini");
```

11 Custom ROOT palette

The `set_color_env()` function is defined near the top of `TPC.C` and is now called in `TPC(...)` immediately after `SetOptStat(0)`, before drawing:

```
gStyle->SetOptStat(0);
set_color_env();
```

Edit its stops and RGB arrays to change the continuous palette; NCont controls the number of colour levels. A later `SetPalette(kViridis)` would override it, Explicit curve/line colours (e.g. voltage overlays) are set separately. Python Matplotlib uses its own cmap settings; the C++ function controls the ROOT workflow.

```
void set_color_env(){
  //*************************************************************
  //code to improve the color palette
  //from the root web page and color codes from:
  //http://ultrahigh.org/2007/08/20/making-pretty-root-color-palettes/
```

```
  //*****************************************************************

  const Int_t NRGBs = 5;
  const Int_t NCont = 255;

  Double_t stops[NRGBs] = { 0.00, 0.34, 0.61, 0.84, 1.00 };
  Double_t red[NRGBs]   = { 0.00, 0.00, 0.87, 1.00, 0.51 };
  Double_t green[NRGBs] = { 0.00, 0.81, 1.00, 0.20, 0.00 };
  Double_t blue[NRGBs]  = { 0.51, 1.00, 0.12, 0.00, 0.00 };
  TColor::CreateGradientColorTable(NRGBs, stops, red, green, blue, NCont);
  gStyle->SetNumberContours(NCont);

  //*****************************************************************


}
```

12 Where to modify the implementation

`analysis/tpc.py`	Commands/menu, common inspection context, 1D/2D drawing, event display styles and exports.
`analysis/detector.py`	ROOT input, wiring, rejection parser/masks, event noise, CSV/axis definitions.
`analysis/voltage.py`	Independent voltage study, normalization, run choice and audits.
`analysis_root/TPC.C`	Public dispatcher (Section 10), drawing/export functions, palette; Sections 8B/8C implement 1D/2D inspection.
`analysis_root/TPCDetector.h`	Native input, mapping, cuts, geometry and noise classification.

For later reconstruction preserve all passing pulses using the hit mask (Python) or raw Hit records and PassHit (ROOT). Mapping supplies pad centres/footprints. DisplayHits/first-pulse reduction is a visualization choice. Drift calibration, reference time, spatial fitting and track output require separate future work.

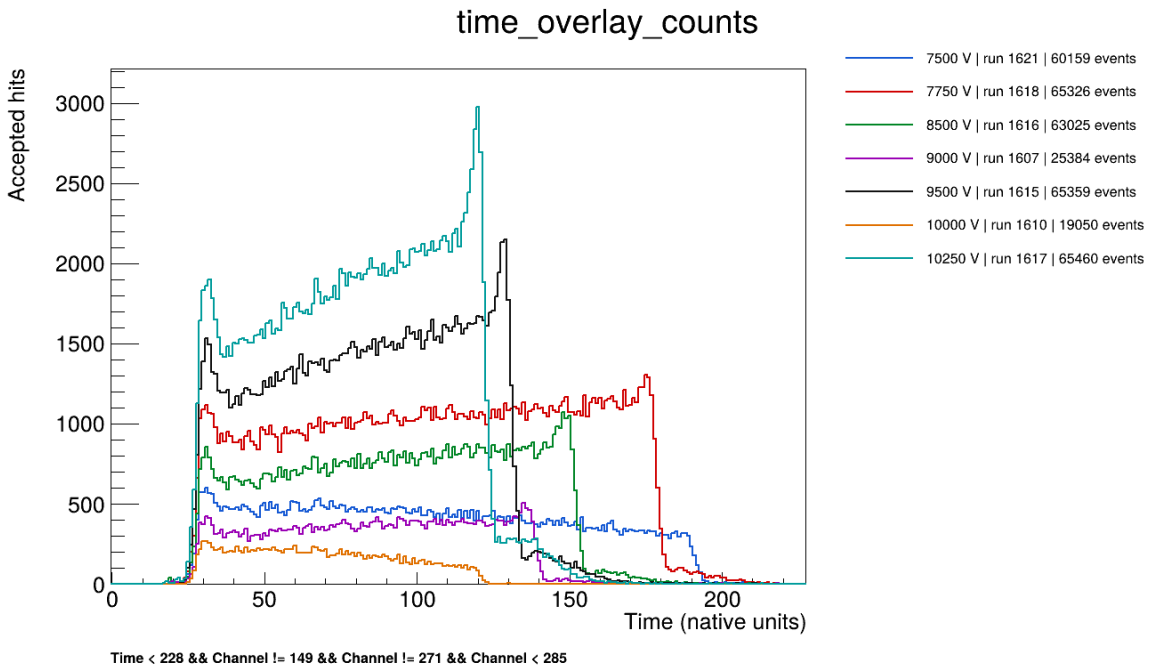


Figure 5: Reference independent voltage-study example: time histograms from runs at different voltages, with detector and study cuts and automatic event vetoes. One eligible run per voltage is selected. This multi-run illustration requires additional data beyond the bundled single-run example.

A Current configuration snapshots

These are the exact files copied at manual build time. Edit the originals in `config/`, not these generated copies.

A.1 Detector cuts

```
# MANUAL REJECTION DATABASE -- shared by Python and C++/ROOT.
# Write the values to REMOVE, not the values to keep.
# Rules within a block are OR-ed: a hit matching ANY reject line is removed.
# Conditions joined by && on ONE line are AND-ed (channel AND time, etc.).
# A run uses [global] and its [run NNNN] block.
# Field names: Event, Channel, Hit, Time, Charge, Pedestal, Blockhead, FADC, Pad.
# Channel is zero-based ROOT electronics; FADC=Channel+1; Pad=-1 is unmapped.
# Rejection removes the HIT ROW from every cleaned plot, including mapped
# Pad plots. Channel==152, FADC==153, and Pad==0 select the same connected pad.
# Write them as alternatives: one rule is sufficient for that pad.
# Operators: == != < <= > >= ; in A..B (inclusive); in A,B,C (list).
# A leading # disables a rule. Text after # is a note, saved in the audit.

[global]
reject = Time >= 228                    # Existing common upper Time cut
# reject = Channel in 271,149             # Existing common excluded ROOT channels

[run 1615]
# Run-specific detector-cleaning rules; commented lines are inactive.

[run 1616]
# Run-specific detector-cleaning rules; commented lines are inactive.
# Active reject lines are applied. Commented examples below are inactive.
reject = Channel == 370               # ROOT370 = fADC371 = large Pad289
reject = Event in 13156..13157          # All hits in these event IDs
# reject = Channel in 40..45            # Inclusive channel range
# reject = Channel == 44 && Time >= 90 && Time < 95  # One channel/time bin region
# reject = Charge < 0                   # Values below a threshold
reject = Charge in 1600..1720               # Values above a threshold
# reject = FADC == 153                  # Same electronics as ROOT Channel 152
reject = Pad == 370                     # Large Pad370 = ROOT564 = fADC565
reject = Pad in 261..263             # Inclusive channel range
#reject = Time > 155
```

A.2 Independent study cuts

```
# INDEPENDENT VOLTAGE STUDY ONLY -- shared by Python and C++/ROOT.
# This file is read only by the voltage-analysis command. Raw inspection
# and event displays do not read it. General detector cleaning belongs in
# run_cuts.ini; the voltage study applies BOTH files to the same hit rows.
# [global] here means every run within THIS STUDY. The same reject syntax
# and optional [run NNNN] blocks are supported as in run_cuts.ini.

[global]
reject = Channel >= 285               # Original comparison electronics range: ROOT channels 0-284
```

A.3 1D binning

```
Branch,BinWidth,Bins,Min,Max,LogY
Event,1,,,,false
Pad,1,410,-0.5,409.5,true
Channel,1,576,-0.5,575.5,true
Hit,1,,,,true
Time,1,,,,false
Charge,1,,,,true
Pedestal,1,,,,true
```

A.4 2D pair definitions

```
Name,X,Y,XBinWidth,XBins,XMin,XMax,YBinWidth,YBins,YMin,YMax,LogZ
Event_vs_Pad,Pad,Event,,,,,64,,,,true
Hit_vs_Pad,Pad,Hit,,,,,,,,,true
Charge_vs_Pad,Pad,Charge,,,,,,,,,true
Charge_vs_Time,Time,Charge,,,,,,,,,true
```

A.5 Python automatic-noise and histogram settings

```
Key,Value,Description
time_max,228,Voltage time-histogram upper bin limit; rejection cuts are in config/run_cuts.ini
noise_channels,288,Distinct channels in an event required for detector-wide noise veto before cuts
module_channels,58,Minimum distinct channels in one module for a module-noise candidate
module_synchronous,50,Distinct channels at identical Time needed unless repeated-hit condition is met
module_hits_per_channel,3,Alternative module condition: total module hits divided by distinct channels
channels_per_module,72,Electronics grouping for module-noise detection; not physical pad rows
readout_modules,8,Number of readout modules in the current setup
excluded_runs,1603,Run excluded from voltage overlays; raw files remain unchanged
```

A.6 ROOT automatic-noise and histogram settings

```
Key,Value,Description
time_max,228,Voltage time-histogram upper bin limit; rejection cuts are in config/run_cuts.ini
noise_channels,288,Distinct channels in an event required for detector-wide noise veto before cuts
module_channels,58,Minimum distinct channels in one module for a module-noise candidate
module_synchronous,50,Distinct channels at identical Time needed unless repeated-hit condition is met
module_hits_per_channel,3,Alternative module condition: total module hits divided by distinct channels
channels_per_module,72,Electronics grouping for module-noise detection; not physical pad rows
readout_modules,8,Number of readout modules in the current setup
excluded_runs,1603,Run excluded from voltage overlays; raw files remain unchanged
```

B Current Python command help

B.1 Inspection

```
usage: tpc.py raw [-h] [--batch] [--dimensions {both,1d,2d} | --1d | --2d |
                  --both] [--plots PLOTS [PLOTS ...]]
                  [--plots-file PLOTS_FILE]
                  [--branches BRANCHES [BRANCHES ...]] [--bins BINS |
                  --bin-width BIN_WIDTH] [--range LOW HIGH] [--log-y |
                  --linear-y] [--binning-file BINNING_FILE] [--apply-cuts]
                  [--input-dir INPUT_DIR] [--output-dir OUTPUT_DIR]
                  [--pad-map PAD_MAP] [--cuts-db CUTS_DB]
                  [--veto-file VETO_FILE]
                  run_number

positional arguments:
  run_number

options:
  -h, --help            show this help message and exit
  --batch               Save outputs without opening a plot window
  --dimensions, --dim {both,1d,2d}
                        Default: both; --branches or --plots alone focuses its
                        own group
  --1d                  Show only 1D
  --2d                  Show only 2D
  --both                Show both groups (default)
  --plots PLOTS [PLOTS ...]
                        Focus on selected 2D plot names
  --plots-file PLOTS_FILE
  --branches BRANCHES [BRANCHES ...]
```

```
                        Focus on individual quantities; default: six-panel
                        mapped summary plus separate Channel diagnostic
  --bins BINS           Override 1D bin count
  --bin-width BIN_WIDTH
                        Override 1D native bin width
  --range LOW HIGH      1D histogram edges, not a data cut
  --log-y
  --linear-y
  --binning-file BINNING_FILE
  --apply-cuts          Compare cleaned data using DB plus automatic vetoes
  --input-dir INPUT_DIR
                        Directory containing ntupleNNNN.root; defaults to
                        data/root/
  --output-dir OUTPUT_DIR
                        Where to save plots; default under results/python/ for
                        the chosen workflow
  --pad-map PAD_MAP
  --cuts-db CUTS_DB     Shared detector global/run rejection database
  --veto-file VETO_FILE
                        Run/event veto CSV; otherwise use live output or
                        bundled auxiliary snapshot
```

B.2 Event display

```
usage: tpc.py event [-h] [--event EVENT] [--mode {2d,3d}] [--save]
                    [--input-dir INPUT_DIR] [--output-dir OUTPUT_DIR]
                    [--pad-map PAD_MAP] [--cuts-db CUTS_DB]
                    [--veto-file VETO_FILE]
                    run_number

positional arguments:
  run_number            Run number, for example 1616

options:
  -h, --help            show this help message and exit
  --event EVENT         Event to start with or export
  --mode {2d,3d}
  --save                Export without opening windows
  --input-dir INPUT_DIR
                        Directory containing ntupleNNNN.root; defaults to
                        data/root/
  --output-dir OUTPUT_DIR
                        Where to save plots; default under results/python/ for
                        the chosen workflow
  --pad-map PAD_MAP
  --cuts-db CUTS_DB     Shared detector global/run rejection database
  --veto-file VETO_FILE
                        Run/event veto CSV; otherwise use live output or
                        bundled auxiliary snapshot
```

B.3 Independent voltage study

```
usage: tpc.py voltage [-h] [--input-dir INPUT_DIR] [--output-dir OUTPUT_DIR]
                      [--voltage-file VOLTAGE_FILE] [--pad-map PAD_MAP]
                      [--cuts-db CUTS_DB] [--study-cuts-db STUDY_CUTS_DB]
                      [--noise-channels NOISE_CHANNELS]
                      [--voltage-override RUN=VOLTS]
                      [--exclude-run EXCLUDE_RUN]
                      [--module-channels MODULE_CHANNELS]
                      [--module-synchronous MODULE_SYNCHRONOUS]

options:
  -h, --help            show this help message and exit
  --input-dir INPUT_DIR
  --output-dir OUTPUT_DIR
  --voltage-file VOLTAGE_FILE
                        Run/voltage table; default checks input-dir, data/,
                        then the bundled auxiliary table
```

```
--pad-map PAD_MAP
--cuts-db CUTS_DB
--study-cuts-db STUDY_CUTS_DB
                      Independent voltage-study restrictions, applied in
                      addition to detector cleaning
--noise-channels NOISE_CHANNELS
                      Reject events with at least this many distinct
                      channels before any cuts
--voltage-override RUN=VOLTS
--exclude-run EXCLUDE_RUN
                      Run excluded from voltage overlays (1603 excluded by
                      default)
--module-channels MODULE_CHANNELS
                      Minimum distinct channels in a 72-channel module
                      (about 80%)
--module-synchronous MODULE_SYNCHRONOUS
                      Required synchronous channels, unless module averages
                      >=3 hits/channel
```